RIVER VALLEY HIGH SCHOOL
YEAR 6 PRELIMINARY EXAMINATION

CANDIDATE NAME										
CLASS	6									
CENTRE NUMBER	S					INDEX NUMBER				

H1 CHEMISTRY **8872/01**

Paper 1 Multiple Choice **26 September 2013**

50 mins

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, class, centre number and index number on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the one you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

This document consists of **15** printed pages.

Section A

For each question there are four possible answers, **A, B, C** and **D**. Choose the **one** you consider to be correct.

1. In an attempt to establish the formula of an oxide of nitrogen, a known volume of the pure N_xO_y gas was mixed with hydrogen and passed over a catalyst at a suitable temperature. 100% conversion of the oxide to ammonia and water was known to have taken place.

2400 cm^3 of nitrogen oxide measured at room temperature and pressure produced 7.20 g of water. The ammonia produced was neutralised by 200 cm^3 of 1.0 mol dm^{-3} of HC*l*.

What is the formula of the nitrogen oxide?

A N_2O **B** N_2O_2

C N_2O_4 **D** NO_2

2. In an experiment, 50.0 cm^3 of a 0.100 mol dm^{-3} solution of a metallic ion, M^{n+}, reacted exactly with 25.0 cm^3 of 0.100 mol dm^{-3} aqueous sodium sulfite.

The half-equation for the oxidation of sulfite ion is shown below.

SO_3^{2-} (aq) + H_2O (l) → SO_4^{2-} (aq) + $2H^+$ (aq) + $2e^-$

If the final oxidation number of the metal in the salt was +3, what would be the original oxidation number of the metal?

A +1 **B** +2 **C** +4 **D** +5

3. Titanium has the electronic structure, $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^2\ 4s^2$.

Which one of the following compounds is **unlikely** to exist?

A TiO **B** $TiCl_3$

C K_3TiF_6 **D** K_2TiO_4

4. Hydrazine, N_2H_4, and hydrogen peroxide, H_2O_2, are both used as rocket propellants because they can produce large volumes of hot gases from a small volume of liquid.

Which of the following statements about these two compounds is correct?

A The bond angle in N_2H_4 is smaller than that in H_2O_2.

B The boiling point of N_2H_4 is lower than that of H_2O_2.

C The N–H bond is shorter than the O–H bond.

D N_2H_4 is a weaker base than H_2O_2.

5. Which type of bonds / attractions is responsible for the intermolecular forces in liquid tetrachloromethane, CCl_4?

A Covalent bonds

B Hydrogen bonds

C Instantaneous dipole-induced dipole interactions

D Permanent dipole-permanent dipole interactions

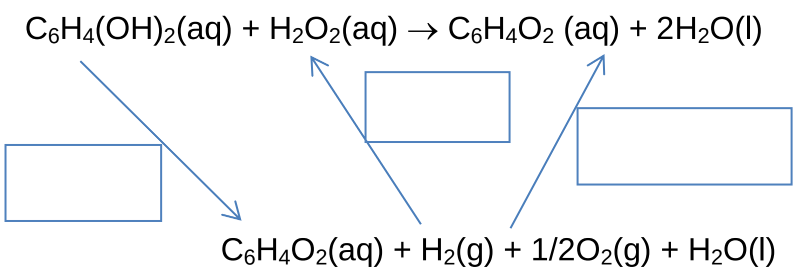
6. When disturbed, bombardier beetles eject a hot noxious chemical spray, from the tip of their abdomen, with a popping sound.

The spray is produced by a reaction between two chemical compounds, hydroquinone ($C_6H_4(OH)_2$) and hydrogen peroxide stored in separate reservoirs in the beetle's abdomen and mixed when needed in a third chamber with water and catalytic enzymes. When threatened, the beetle squeezes some fluid from the inner compartment into the outer compartment, where a reaction takes place

$$C_6H_4(OH)_2(aq) + H_2O_2(aq) \rightarrow C_6H_4O_2\ (aq) + 2H_2O(l) \qquad \Delta H_{rxn} = ?$$

To calculate the enthalpy change of reaction, an energy cycle is drawn.
The energy cycle and additional information are as shown below.

$C_6H_4(OH)_2(aq) \rightarrow C_6H_4O_2(aq) + H_2(g)$ $\quad \Delta H_1 = +177.4$ kJ mol^{-1} $O_2(g) + 2H_2O(l) \rightarrow 2H_2O_2(aq)$ $\quad \Delta H_2 = +189.1$ kJ mol^{-1} Enthalpy change of formation of water = –285.8 kJ mol^{-1}



What is the value of the enthalpy change of reaction?

A –203 kJ mol^{-1} **B** –489 kJ mol^{-1}

C –13.8 kJ mol^{-1} **D** –300 kJ mol^{-1}

7. The lattice energies of rubidium fluoride, RbF, and caesium chloride, Cs*Cl*, are –760 kJ mol^{-1} and –650 kJ mol^{-1} respectively.

Which value is likely to be the lattice energy of caesium fluoride, CsF, in kJ mol^{-1}?

A –460 kJ mol^{-1}

B –550 kJ mol^{-1}

C –680 kJ mol^{-1}

D –800 kJ mol^{-1}

8. ΔH_f of H_2O is –285 kJ mol^{-1}

ΔH_f of CO_2 is –394 kJ mol^{-1}

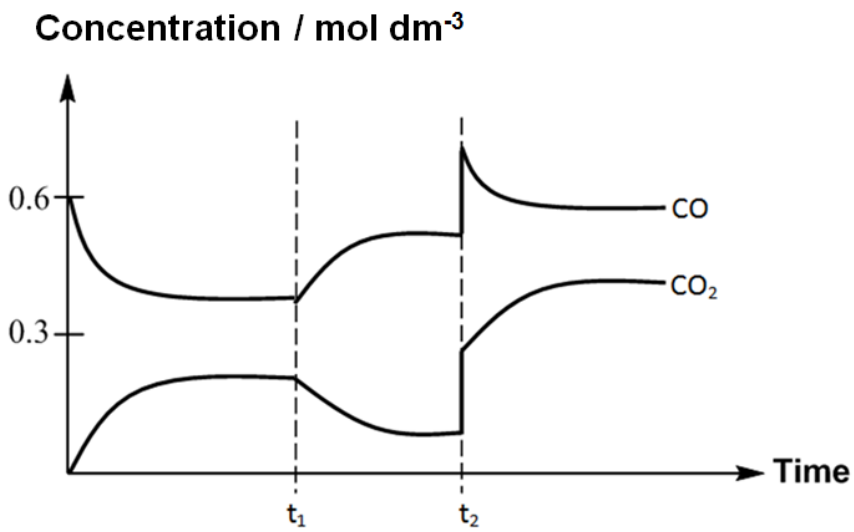
What is the ratio of heat generated by hydrogen to that of carbon if the same mass of each substance is burned completely?

A 0.4 **B** 0.7 **C** 3.4 **D** 4.3

9. At a temperature T K, 0.60 mol dm^{-3} of CO and 0.30 mol dm^{-3} of O_2 were introduced into a 5 dm^3 vessel and allowed to reach equilibrium.

$$2\,CO\,(g) + O_2\,(g) \rightleftharpoons 2CO_2\,(g) \qquad \Delta H < 0$$

The graph below shows the changes in the concentration of CO and CO_2 in the system with time. A change was made to the system at time, t_1 and t_2.



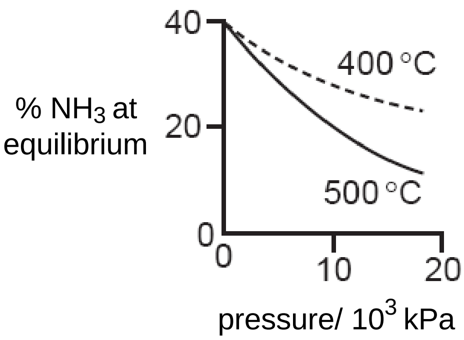
What were the changes made at time, t_1 and t_2?

	t_1	t_2
A	A catalyst was added	Volume of the system is increased
B	More CO_2 was added	An inert gas was added at constant volume
C	The temperature was decreased	More O_2 was added
D	The temperature was increased	Volume of the system is decreased

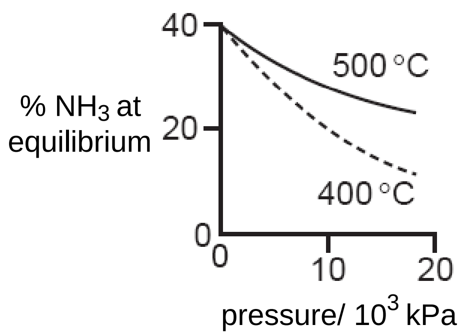
10. The percentage of ammonia obtainable, if equilibrium was established during the Haber process, is plotted against the operating pressure for two temperatures, 400 °C and 500 °C.

Which diagram correctly represents the two graphs?

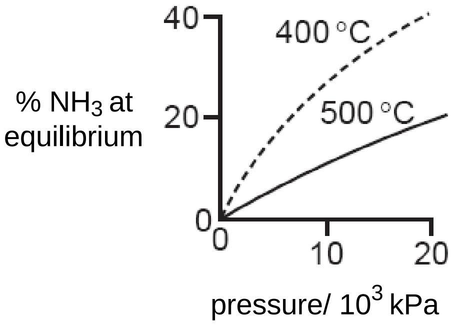
A



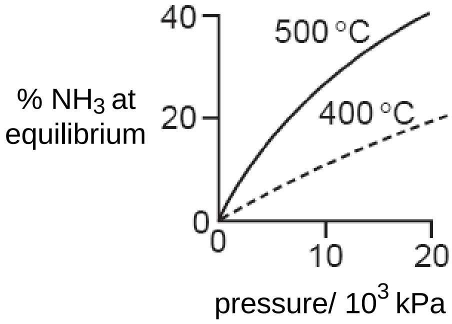
B



C



D



11. Water dissociates into H^+ and OH^- as shown.

$$H_2O \rightleftharpoons H^+ + OH^-$$

At 25 °C, the equilibrium $[H^+]$ is 10^{-7} mol dm^{-3}; $[H_2O]$ = 55.6 mol dm^{-3}.

What is the order of increasing numerical value of pH, pK_a and pK_w for this equilibrium at this temperature?

	smallest		*largest*
A	pH	pK_w	pK_a
B	pH	pK_a	pK_w
C	pK_w	pK_a	pH
D	pK_a	pK_w	pH

12. When magnesium carbonate powder is added to a solution of aqueous ethanoic acid, effervescence is observed.

Which of the following will **not** change the rate of effervescence?

A Using magnesium carbonate pellet instead of magnesium carbonate powder.

B Using hydrochloric acid instead of ethanoic acid.

C Increasing the temperature.

D Increasing the pressure.

13. Two solutions were prepared by dissolving a chloride and an oxide of elements in the third period of the Periodic Table in separate portions of water.

Both solutions prepared can be used to dissolve Al_2O_3 but only one can be used to dissolve SiO_2.

Which of the following could be the chloride and the oxide used?

A PCl_5 and Na_2O

B $NaCl$ and SO_3

C $MgCl_2$ and MgO

D $SiCl_4$ and P_4O_{10}

14. **X**, **Y** and **Z** are consecutive elements in Period 3 of the Periodic Table. **Y** has the highest first ionisation energy and the lowest melting point among the three elements.

Which of the following could be **X**, **Y** and **Z**?

A sodium, magnesium, aluminium

B magnesium, aluminium, silicon

C aluminium, silicon, phosphorus

D silicon, phosphorus, sulfur

15. An enzyme, found in the stomach, operates at maximum efficiency when placed in an aqueous solution buffered at pH 5.

Which combination of substances, when dissolved in 10 dm^3 of water, would give the necessary buffer solution?

A 1 mol HC*l* and 1 mol of CH_3COOH

B 1 mol CH_3COOH and 1 mol of CH_3COONa

C 1 mol HC*l* and 1 mol of CH_3COONa

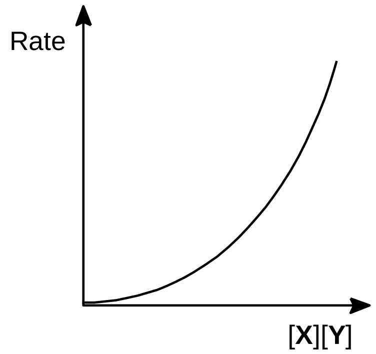
D 1 mol of CH_3COONH_4

16. The reaction between **X** and **Y** is an overall second–order reaction.

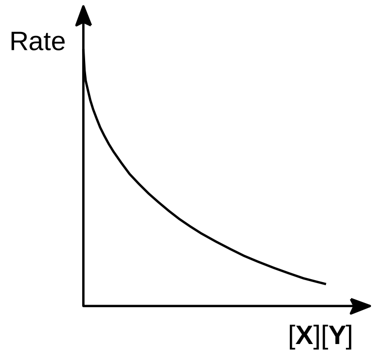
$$\mathbf{X} + \mathbf{Y} \rightarrow \mathbf{Z}$$

The rate of this reaction is plotted against the product [**X**][**Y**] of the concentrations of **X** and **Y**. Which graph would be obtained?

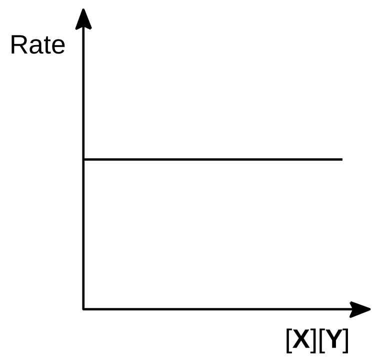
A



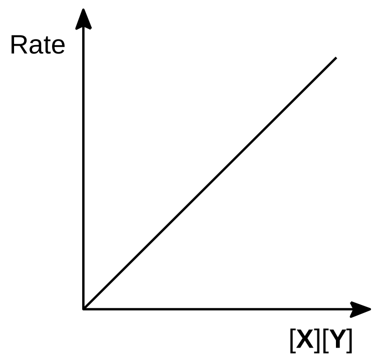
B



C



D



17. Assuming that only mono-chlorination takes place, which of the following will be the correct combination of the products obtained when excess 2,3-dimethylbutane is reacted with chlorine in the presence of *uv* light?

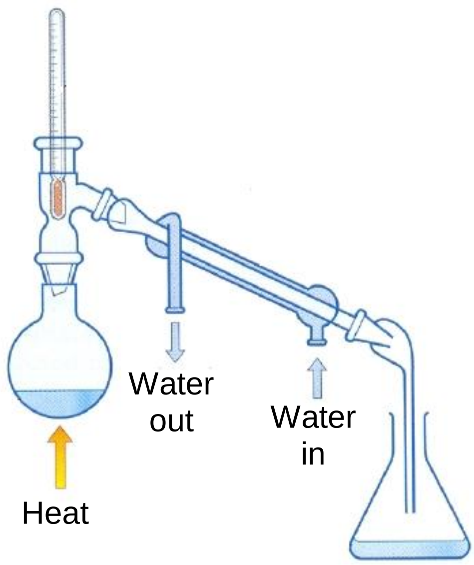
A 2 possible products in the ratio 6:1

B 3 possible products in the ratio 3:3:1

C 3 possible products in the ratio 6:3:1

D 4 possible products in the ratio 6:3:3:2

18. Which of the following **cannot** be prepared by the apparatus set-up as shown?



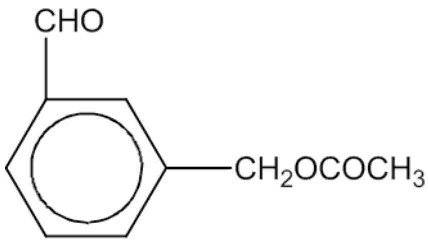
A 1,2-dibromopropene by using bromine and propene

B Ethyl ethanoate by using ethanol and concentrated sulfuric acid

C Bromoethane by using sodium bromide and concentrated sulfuric acid

D Propanone by using propanol, sodium dichromate(VI) and sulfuric acid

19. Which of the following reagents will **not** show a reaction when the reagent is added to compound **X**?



Compound **X**

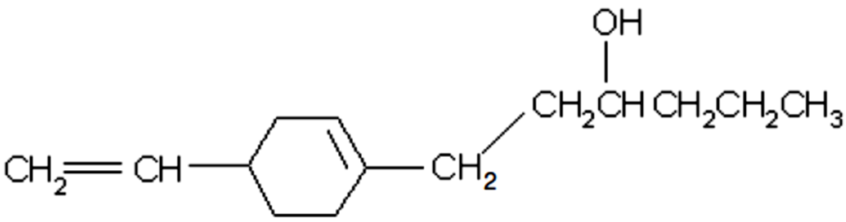
A Acidified potassium manganate(VII) solution

B Alkaline silver nitrate and ammonia solution

C Alkaline copper(II) sulfate solution

D Na metal

20. What is the total number of isomers that can be formed from the product of the reaction of the following compound with excess concentrated H_2SO_4 at $170^{o}C$?



A 2

B 4

C 8

D 16

21. ^{18}O is an isotope of oxygen.

When propyl ethanoate is hydrolysed with dilute hydrochloric acid in the presence of $H_2{}^{18}O$, a mixture of two products is formed. Which of the following pairs gives the correct structures of the two products?

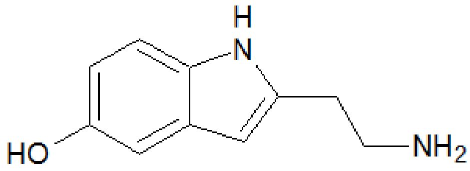
A CH_3COOH and $CH_3CH_2CH_2{}^{18}OH$

B $CH_3CO^{18}OH$ and $CH_3CH_2CH_2OH$

C CH_3CH_2COOH and $CH_3CH_2{}^{18}OH$

D $CH_3CH_2CO^{18}OH$ and CH_3CH_2OH

22. Serotonin is a monoamine neurotransmitter that contributes to feelings of well-being and happiness.



serotonin

How many sigma (σ) and pi (π) bonds does serotonin have?

A 20σ and 4π

B 22σ and 4π

C 26σ and 4π

D 28σ and 4π

23. Which of the following will **not** be produced when 1-bromopropane is heated with ethanolic sodium hydroxide?

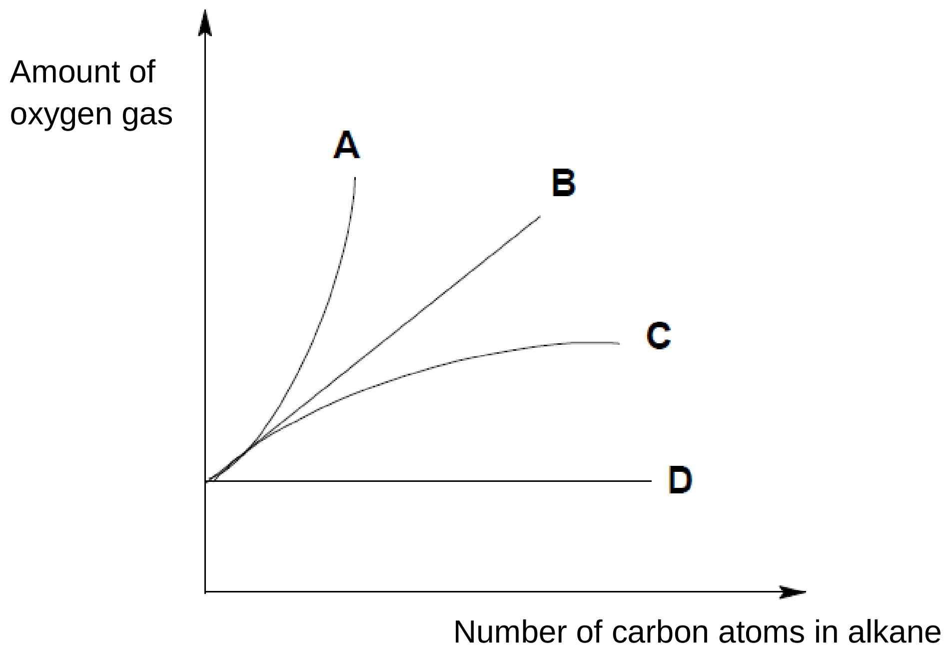
A $CH_3CH{=}CH_2$

B $CH_3CH_2CH_2OH$

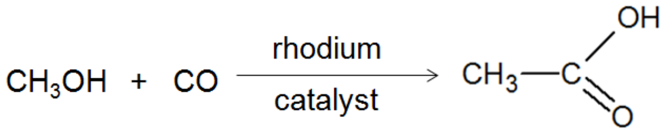
C $CH_3CH_2CH_2ONa$

D $CH_3CH_2CH_2OCH_2CH_3$

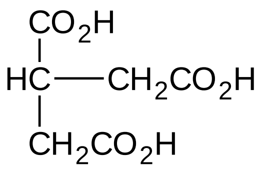
24. Which line on the graph shows the relationship between the number of carbon atoms in an alkane and the amount of oxygen gas needed for complete combustion of the alkane?

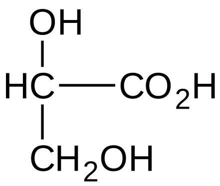


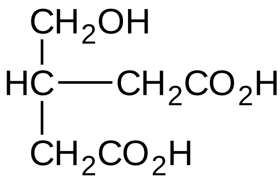
25. Ethanoic acid is prepared industrially by the direct carbonylation of methanol using a rhodium catalyst.

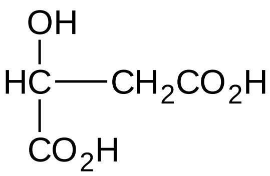


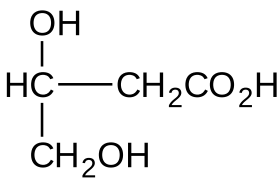
Which compound can be expected to produce the following product by this method?



A 

B 

C 

D 

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct.

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

26. The electronic configurations of elements **X**, **Y** and **Z** are as follows:

X: [Ar] $3d^{10}4s^{2}4p^{1}$

Y: [Ar] $3d^{10}4s^{2}4p^{4}$

Z: [Ar] $3d^{10}4s^{1}$

Which of the following statement(s) is / are true?

1 $\mathbf{X}^{3+}$ shows the greatest deflection towards the negatively charged plate with constant electric field compared to $\mathbf{Y}^{+}$ and $\mathbf{Z}^{2+}$.

2 The first ionization energy of **X** is lower than both **Z** and **Y**.

3 Each of the $\mathbf{X}^{2+}$, $\mathbf{Y}^{2+}$ and $\mathbf{Z}^{2+}$ ions contain only one unpaired electron.

27. Which of the following trends concerning Period 3 elements (sodium to sulphur) are true?

1 There is a change from metallic to non-metallic behaviour.

2 Their compounds show an increase in the maximum oxidation number of the element across the period.

3 Their compounds become less ionic and more covalent across the period.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

28. Given the following reaction:

$$\mathbf{P}(g) + \mathbf{Q}(g) \rightleftharpoons \mathbf{X}(g) + \mathbf{Y}(g) \qquad \Delta H = \text{Negative}$$

Forward reaction: $(\text{Rate})_f = k_f[\mathbf{P}][\mathbf{Q}]$

Backward reaction: $(\text{Rate})_b = k_b[\mathbf{X}][\mathbf{Y}]$

Which of the following is correct?

1 Increasing the pressure will not cause k_f and k_b values to change and the equilibrium position remains the same.

2 Increasing the temperature will only increase the value of k_b, which causes the equilibrium position to shift to the left.

3 Decreasing the pressure will not cause $(\text{Rate})_f$ and $(\text{Rate})_b$ values to change and the equilibrium position remains the same.

29. Deuterium, D (^{2}H), is a heavy isotope of hydrogen. Which of the following reactions will incorporate D in the final product?

1 $CH_3COOCH_2CH_3$ + hot aqueous NaOD

2 $CH_2{=}CH_2 + Br_2$ (in D_2O)

3 CH_3COCH_3 + DCN

30. In which of the following pairs of organic compounds is the compound on the left more volatile than the one on the right?

1 Propylamine and propan-1-ol

2 Cyclohexylamine and aminoethanoic acid

3 Phenylethanol and phenylethanoate

End of paper

Suggested Answers

1	**C**	11	**A**	21	**B**
2	**C**	12	**D**	22	**C**
3	**D**	13	**A**	23	**C**
4	**B**	14	**D**	24	**B**
5	**C**	15	**B**	25	**D**
6	**A**	16	**D**	26	**B**
7	**C**	17	**A**	27	**A**
8	**D**	18	**A**	28	**D**
9	**D**	19	**C**	29	**C**
10	**C**	20	**B**	30	**B**